

1. Differentiability implies continuity

Theorem

If a function f is differentiable at a then f is continuous at a .

(So the theorem says if $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = f'(a)$ (a finite number) then $\lim_{x \rightarrow a} f(x) = f(a)$)

Proof

$$\begin{aligned} \lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} (f(x) - f(a) + f(a)) && \text{Addition or subtraction of the same term} \\ &= \lim_{x \rightarrow a} (f(x) - f(a)) + \lim_{x \rightarrow a} f(a) && \text{Limit law of addition} \\ &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} (x - a) + f(a) && \text{Multiplication and division by the same term} \\ &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \cdot \lim_{x \rightarrow a} (x - a) + f(a) && \text{Limit law of multiplication} \\ &= f'(a) \cdot 0 + f(a) && f \text{ is differentiable} \\ &= f(a) \end{aligned}$$

So $\lim_{x \rightarrow a} f(x) = f(a)$ and therefore f is continuous in $x = a$.

Note: If f is continuous in a then f is not necessarily differentiable in a .

Counter example: $f(x) = |x|$ is continuous in 0 but not differentiable in 0.

2. (i) A positive derivative indicates that the function is increasing.

Definition of an increasing function:

A function f is increasing on an interval I if $f(x_2) > f(x_1)$ for any $x_1, x_2 \in I$ such that $x_2 > x_1$.

Theorem: If $f' > 0$ on an interval I then f is increasing on the interval.

Proof: Pick any two points $x_1, x_2 \in I$ such that $x_2 > x_1$.

Consider the function f on the interval $[x_1, x_2]$.

1. f is continuous on $[x_1, x_2]$ (f is differentiable on I)
2. f is differentiable on (x_1, x_2) (f is differentiable on I)

So, according to the Mean Value Theorem a $c \in (x_1, x_2)$ exists such that

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

But $f'(c) > 0$ and $x_2 - x_1 > 0$, therefore $f(x_2) - f(x_1) > 0$.

It follows that $f(x_2) > f(x_1)$ and therefore f is increasing on I .

2 (ii) A negative derivative indicates that the function is decreasing.

Definition of an decreasing function:

A function f is decreasing on an interval I if $f(x_2) < f(x_1)$ for any $x_1, x_2 \in I$ such that $x_2 > x_1$.

Theorem: If $f' < 0$ on an interval I then f is decreasing on the interval.

The proof of this theorem is similar to that given above.

3. Derivative of $y = \sin x$

$$\begin{aligned}\frac{d}{dx} \sin x &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} \\&= \lim_{h \rightarrow 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} \\&= \lim_{h \rightarrow 0} \frac{\sin x (\cos h - 1) + \cos x \sin h}{h} \\&= \sin x \lim_{h \rightarrow 0} \frac{(\cos h - 1)}{h} + \cos x \lim_{h \rightarrow 0} \frac{\sin h}{h} \\&= \cos x \quad \text{because } \lim_{h \rightarrow 0} \frac{(\cos h - 1)}{h} = 0 \text{ and } \lim_{h \rightarrow 0} \frac{\sin h}{h} = 1\end{aligned}$$

We show why $\lim_{h \rightarrow 0} \frac{(\cos h - 1)}{h} = 0$:

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{(\cos h - 1)}{h} &= \lim_{h \rightarrow 0} \frac{(\cos h - 1)}{h} \cdot \frac{\cos h + 1}{\cos h + 1} \\&= \lim_{h \rightarrow 0} \frac{(\cos^2 h - 1)}{h(\cos h + 1)} \\&= \lim_{h \rightarrow 0} \frac{-\sin^2 h}{h(\cos h + 1)} \\&= \lim_{h \rightarrow 0} \frac{\sin h}{h} \cdot \lim_{h \rightarrow 0} \sin h \cdot \lim_{h \rightarrow 0} \frac{-1}{(\cos h + 1)} = 1 \times 0 \times \left(-\frac{1}{2}\right) = 0\end{aligned}$$

4. Other trig derivatives

$$\begin{aligned}\frac{d}{dx} (\tan x) &= \frac{d}{dx} \left(\frac{\sin x}{\cos x} \right) \\&= \frac{\cos x \cos x + \sin x \sin x}{(\cos x)^2} \\&= \frac{1}{\cos^2 x} \\&= \sec^2 x\end{aligned}$$

$$\begin{aligned}\frac{d}{dx} (\operatorname{cosec} x) &= \frac{d}{dx} \frac{1}{\sin x} \\&= \frac{-\cos x}{\sin^2 x} \\&= -\frac{\cos x}{\sin x} \cdot \frac{1}{\sin x} \\&= -\operatorname{cosec} x \cdot \cot x\end{aligned}$$

Derivatives of $\cot x$ and $\sec x$ follow similarly.